Layouts Made in Diagram Editors Are Held by Alignment and Flow, Not by Edge Length or Stress: A Per-Term Stationarity Test Before Fitting a Layout Energy

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7 September 2026

Abstract

Fitting the weights of a layout energy to the coordinates of layouts people made assumes each layout is a local minimum of some weighting of the criteria. Per criterion, move each box 0.02 of the drawing width in sixteen directions and count the boxes no move improves, against a converged layout and a random one. On 853 working diagrams of 15 to 40 boxes from three editors, overlap holds every box. Uniform edge length and stress hold none, there and in 136 laboratory drawings and 1,072 figures from the Graph Drawing proceedings. A stress minimiser's own layouts are held at 1.00, and the hand layouts carry five to twelve times the minimiser's stress. A weight of 0.001 on either term beside crossings and overlap holds under 0.015 of the boxes. Two criteria the base energies omit do hold the working diagrams. Alignment: 58, 36 and 81 per cent of boxes sit in a row or column of three or more, against 4 to 7 by chance. Flow: 0.59 to 0.86 of boxes are held, against 0.28 to 0.41. Person or editor, the coordinates cannot say which aligned them.

1 Introduction

A layout program places the boxes of a diagram by minimising an *energy*, a weighted sum of quantities the field has agreed are bad [1, 2, 3, 4]. A term that carries a criterion in the energy is said below to *price* it. Where an objective is fitted to what people did, the standard estimator takes the demonstration as a stationary point of the unknown objective [5]. Inverse optimisation minimises the residual of the optimality conditions at the observed decision [6, 7]. Inverse optimal control of manipulation [8] and of arm movement [9] assumes the demonstrations satisfy those conditions, and inverse reinforcement learning assumes the demonstrated policy optimal [10] or locally so [11]. In layout the published fit of this kind is O'Donovan, Agarwala and Hertzmann's energy for single-page designs, each example taken as optimal [12]; a reward fitted to expert chip placements rests on the same premise [13]. Graph drawing has learned weights from preferences, selections and labels since 1994 [14, 15, 16, 17], which needs only that an accepted layout rank well, and has imitated the coordinates of algorithm output [18, 19]. No published graph-drawing method fits an energy to human coordinates. The corpus to do it on now exists, 4,890 figures from this field's proceedings [21], and Flud [22] already continues a descent from crowd layouts of pathway diagrams. This paper is the check to run before that fit.

*Independent researcher. Code, parsed corpora, every table's script and its committed output: `https://github.com/Anode1/cjitter`, directory `example/diagrams` at commit `c747a80`; `results/manifest.txt` records the platform, tool versions, seeds and archive hashes of the 23 August 2026 battery, and the later runs (the BPMN random 300, the routes, the flow term, the A3 refit) sit each beside its script with the date in its header. `data/PROVENANCE.md` gives each corpus's source, version, hash and licence. The directional test is deterministic; the descents are deterministic from their seed.

corpus	term	hand	`neato`	`dot`	ELK	random
WikiPathways	crossings	1.00 [1.00, 1.00] (0.000)	1.00 (0.000)	1.00 (0.029)	1.00 (0.036)	0.67
	overlap	1.00 [1.00, 1.00] (0.000)	0.60 (0.030)	1.00 (0.000)	1.00 (0.000)	0.67
	length	0.00 [0.00, 0.00] (0.219)	0.24 (0.011)	0.00 (0.241)	0.00 (0.127)	0.00 (0.193)
	stress	0.00 [0.00, 0.00] (0.217)	1.00 (0.042)	0.00 (0.201)	0.00 (0.241)	0.00 (0.435)
	orthogonality	0.19 [0.18, 0.21] (0.157)	0.00 (0.280)	0.19 (0.244)	0.19 (0.158)	0.00
	alignment A1	0.52 [0.48, 0.54] (0.003)	0.06 (0.011)	1.00 (0.000)	0.57 (0.001)	0.07
	gridiness	0.87 [0.84, 0.89] (0.424)	0.78 (1.000)	1.00 (0.125)	0.76 (0.200)	(0.96)
	node–edge	0.88 [0.86, 0.90] (0.004)	0.89 (0.000)	0.78 (0.011)	0.74 (0.016)	0.12
	flow	0.59 [0.56, 0.62] (0.131)	0.50 (0.211)	1.00 (0.000)	1.00 (0.000)	0.41
Reactome	crossings	1.00 [1.00, 1.00] (0.000)	1.00 (0.000)	1.00 (0.056)	1.00 (0.066)	0.61
	overlap	1.00 [1.00, 1.00] (0.000)	0.76 (0.022)	1.00 (0.000)	1.00 (0.000)	0.67
	length	0.00 [0.00, 0.00] (0.234)	0.35 (0.005)	0.00 (0.232)	0.00 (0.205)	0.00 (0.188)
	stress	0.00 [0.00, 0.00] (0.212)	1.00 (0.034)	0.00 (0.177)	0.00 (0.197)	0.00 (0.469)
	orthogonality	0.17 [0.15, 0.18] (0.193)	0.00 (0.277)	0.15 (0.256)	0.12 (0.203)	0.00
	alignment A1	0.21 [0.19, 0.24] (0.004)	0.06 (0.009)	1.00 (0.000)	0.42 (0.002)	0.06
	gridiness	0.65 [0.62, 0.68] (0.640)	0.77 (1.000)	0.95 (0.176)	0.74 (0.306)	(0.93)
	node–edge	1.00 [0.94, 1.00] (0.000)	0.93 (0.000)	0.80 (0.007)	0.75 (0.021)	0.11
	flow	0.62 [0.60, 0.67] (0.096)	0.45 (0.217)	1.00 (0.000)	1.00 (0.000)	0.36
BPMN	crossings	1.00 [1.00, 1.00] (0.000)	1.00 (0.000)	1.00 (0.000)	1.00 (0.000)	0.62
	overlap	1.00 [1.00, 1.00] (0.000)	0.41 (0.071)	1.00 (0.000)	1.00 (0.000)	0.60
	length	0.00 [0.00, 0.00] (0.284)	0.40 (0.005)	0.00 (0.222)	0.00 (0.250)	0.00 (0.215)
	stress	0.00 [0.00, 0.00] (0.197)	1.00 (0.016)	0.00 (0.152)	0.00 (0.175)	0.00 (0.507)
	orthogonality	0.73 [0.71, 0.75] (0.011)	0.00 (0.277)	0.27 (0.169)	0.25 (0.145)	0.00
	alignment A1	0.85 [0.82, 0.89] (0.001)	0.06 (0.011)	1.00 (0.000)	0.59 (0.001)	0.06
	gridiness	0.95 [0.94, 0.96] (0.188)	0.79 (1.000)	0.81 (0.238)	0.67 (0.279)	(0.95)
	node–edge	1.00 [1.00, 1.00] (0.000)	1.00 (0.000)	0.86 (0.001)	0.79 (0.011)	0.11
	flow	0.86 [0.83, 0.88] (0.041)	0.62 (0.120)	1.00 (0.000)	1.00 (0.000)	0.28

Table 1: Median $q_{0.02}$ over diagrams, the fraction of a diagram's boxes no move of 0.02 of the width in sixteen directions improves, per criterion alone: the hand layout with its 95% bootstrap interval, `neato`, `dot` and ELK layouts of the same graphs, and a uniform random layout, one per diagram, on chord edges. In parentheses the median value of the term at the layout, 0 when satisfied; for gridiness, a step whose q reads flatness, the value is the quantity compared, and the random column gives its value. Terms and controls are defined in Section 2, the corpora in Section 3.

Move each box of a drawing a small distance in sixteen directions with the rest fixed; call the box *held* if no move lowers a criterion; report the fraction of boxes held, q, per criterion. Two controls: a converged layout, which checks the test's power, and a random layout, which gives the chance line. Run before any weight is fitted, the count sorts the terms into satisfied, inert and violated at the demonstrations, and a term that holds the demonstrations while the fitted energy holds none is a missing feature found without fitting. The fitting literature, when it measures the demonstrations' distance from optimality, does so inside the fit and in aggregate [7]. Table 1 is the result on three working corpora, the layouts people made in their editors (hand layouts below), against three tool controls and a random null. Three things are new: the per-term held count at human coordinates with those controls, the drop to under 0.015 held at any weight on a distance term, and the profile, the vector of q per term, which hand layouts share with the layered engines.

2 The test

The energy. A layout of a graph with N nodes is a point $x \in \mathbb{R}^{2N}$, two coordinates per node, the boxes' sizes kept fixed. An aesthetic energy is

$$E_w(x) = \sum_{k=1}^{K} w_k \, T_k(x), \qquad w_k \geq 0, \tag{1}$$

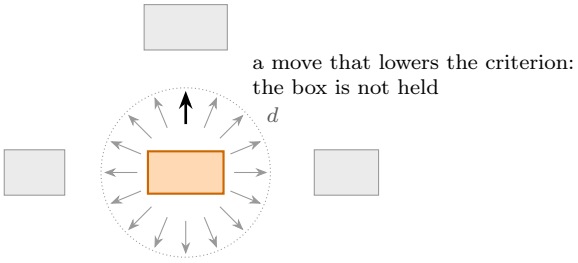


Figure 1: The test, one box and one criterion. The box is moved a distance d, two per cent of the drawing's width, in each of sixteen exact directions, and the criterion is recomputed. If some move lowers it (the dark arrow), the box is not held; if no move lowers it, the box is *held*.

where each term T_k is a quantity to be made small. Multicriteria layout is the choice of the T_k and the w_k [2, 3].

Held. A point x is a local minimum of E if $E(y) \geq E(x)$ for every y within some distance of x. Two of the base terms are not differentiable, crossings being a count and overlap having corners where boxes touch, so the test uses no gradient. Fix a radius d and a set V of sixteen equally spaced unit directions. For node i and direction v let $x + d\,v^{(i)}$ be the layout with node i moved by d along v and every other node fixed (Figure 1). Node i is *held* at radius d if

$$E(x + d\,v^{(i)}) \geq E(x) \quad \text{for every } v \in V, \tag{2}$$

and the estimand is

$$q_d(x) = \frac{1}{N}\,\#\{\, i : \text{node } i \text{ is held at radius } d \,\}. \tag{3}$$

$q_d = 1$ says the layout is a local minimum of E over single-node moves of length d, a weaker property than a minimum over joint moves at every small radius, and the one meant wherever this paper says minimum. For smooth E and small d it says $\nabla E(x) = 0$ to within the resolution of V. A person in a diagram editor moves one box at a time, so single-node moves are the editor's own move set. Equation (2) compares two energies, so it is defined for every term. The primary radius is $d = 0.02$ of the drawing width, the larger extent of its bounding box (width below). A move that lowers the energy by less than 10^{-12} of its value counts as no change. The test is deterministic: no seed, no budget.

Three quantities. The tables report the *median* q over diagrams, called the *hold* of a term below. The fit and the weight sweep report the *pooled* q, the fraction of all boxes in a corpus held. Pooled q is below the median wherever diagrams pile up at 1: in the sweep's alignment row WikiPathways reads 0.52 by the median and 0.506 pooled. The third quantity is the *value* of a term at the layout, 0 when satisfied. A held fraction means different things for terms of different shape. For a smooth term near a minimum with gradient g and curvature λ along it, a node is held when $|g| < d\lambda/2$: q reads how near the minimum is. For a count (crossings) or a step function of position (gridiness), a node is held wherever no move of length d changes the term: q reads how much of the layout the term has stopped acting on, so the value is compared instead. For a term with a corner at exact coincidence (the alignment term A1), a node is held when it sits within $0.19\,d$ of sharing a coordinate with another node (Supplement S2) and is free to slide along that row or column. Here q counts boxes pinned in one coordinate, at a resolution the direction set fixes.

The inverse problem. Given layouts $x^{(1)}, \ldots, x^{(G)}$ that people accepted, the first-order fit asks for the weights w under which they are nearest to stationary, minimising the residual

of the optimality conditions at the demonstrations [6, 8]. With the directional differences $D_{iv}T_k = T_k(x + d\,v^{(i)}) - T_k(x)$ in hand, node i is held under w exactly when $\sum_k w_k D_{iv}T_k \geq 0$ for all v, a set of linear inequalities in w. The residual is the smallest total violation over w on the simplex (the normalisation that keeps the trivial vertex feasible). Finding it is a linear program, and it is zero only if some weighting holds every box. Two properties of that program decide what the fit can return. A term with $D_{iv}T_k < 0$ for some v at almost every node can take positive weight only where another term rises in that direction. Under the base energy nothing else rises there, and the sweep of Section 4 measures how much weight such a term can carry. And a term every layout satisfies exactly, its value zero so that $D_{iv}T_k \geq 0$ everywhere, is a zero-residual vertex on its own. The program returns all weight on it and learns nothing. That is the degeneracy Ng and Russell [10] state for inverse reinforcement learning, where a zero reward makes every policy optimal. Aswani, Shen and Siddiq [25] prove the residual estimators inconsistent under noisy demonstrations. The margin [23] and maximum-entropy [24] repairs read a gap between the demonstration and its alternatives, a different quantity, and Section 7 says what they would return here.

The reference length. The length and stress terms compare distances with a reference L. Fixed at the layout's median edge length, L is not the scale at which the term is least, so a true minimum still has a scaling move that lowers it: `neato`'s converged stress layouts read $q = 0.07$ to 0.12 under their own criterion, and that observation fixed the rule. L is fixed before any move at the length that makes the term least for the layout as loaded: $\sum \ell^2 / \sum \ell$ over edges for length, and the same over $\|x_i - x_j\|/\delta_{ij}$ for stress. This is the scale-normalised stress of Smelser, Miller and Kobourov [26] and of Ahmed et al. [27]. The converged layouts are then held at 1.00; the hand verdicts are 0.00 to 0.05 under every rule tried (Supplement S3).

The controls. The *converged reference*: q_d on a layout a descent has converged on must be near one, or the test lacks power for that energy. `neato`, an independent minimiser of stress, is the power control: its layouts are held at 1.00 under stress and the length term is of the same family. The end point of an uncapped descent from each hand layout is the reference for every summed energy (median q 1.00 in every corpus, Supplement S4). That climber proposes the test's own single-box moves, so this reference certifies the code and not the power. The *tool controls*: the same graphs with the same box sizes and edge directions laid out by `neato` (stress, `-Gstart=1`, boxes left where they overlap), `prism` (`neato` followed by its overlap removal), `dot` (layered) and ELK layered 0.12.0 [28]. ELK runs as `elkjs` at the library's defaults (node placement Brandes–Köpf with edge straightening, direction right, seed 1), not as any of the editors that offer it configure it. Graphviz is the Ubuntu 24.04 package 2.42.2, which reports itself as 2.43.0, built with GTS so that `-Goverlap=false` is `prism`. The tools receive the graph without its containers and with undirected edges in stored order. Their routes are discarded, so crossings, orthogonality and node– edge separation are read on chords in the tool columns and on the stored routes in the hand column. Supplement S3 gives the hand column on chords. A check in the binary confirms that each control lays out the same graphs; the command lines are `data/make_corpus.py` and `data/elk_layout.py`. The *null*: a uniform random layout of the same graph in the same bounding box, whose q is the chance line, and for the alignment term a random layout snapped to the same grid as the drawing.

Terms. Table 2. The base energy is $T_C + T_O + T_L$ at weights $(1, 1, 1)$, three terms of the family that runs from Davidson and Harel's simulated annealing [2] to the gradient-descent layouts of Ahmed et al. [30]; stress [31, 32], which force-directed tools and `neato` minimise, is the second base energy. T_R, T_A and T_N are the candidates for what the base energies miss. A1 has a corner at exact alignment and A3 has none; A1 is the primary term, A3 the sensitivity check (the binary

term	definition	zero when
T_C crossings	proper crossings between segments of two routes with no shared endpoint, $/m$	planar
T_O overlap	overlap area of boxes / total box area	no overlap
T_L length uniformity	mean over edges of $((\ell - L)/L)^2$	all edges of length L
T_S stress	mean over node pairs of $((\lVert x_i - x_j \rVert - L\,\delta_{ij})/(L\,\delta_{ij}))^2$, δ_{ij} the graph distance	distances proportional
T_R orthogonality	mean over edges of the length-weighted mean over its segments of $\min(\lvert\Delta x\rvert, \lvert\Delta y\rvert)/(\lvert\Delta x\rvert + \lvert\Delta y\rvert)$	routes axis-aligned
T_A alignment	A1: mean over nodes of the distance to the nearest other node along whichever axis is nearer; A3: mean over nodes of $1/(1 + \sum_{j\neq i} e^{-(\Delta x_{ij}/s)^2} + e^{-(\Delta y_{ij}/s)^2})$, $s = 0.02$, smooth	every node in a row or column
gridiness	$1-$ fraction of nodes in an alignment of ≥ 3 (tolerance 0.005), HOLA's definition [29], a step	every node aligned
T_N node–edge	squared shortfall of the node-to-foreign-route distance below $(w + h)/4$	none within
T_F flow	mean over directed edges of the backward component along the reading direction u, over length	no edge runs back

Table 2: The terms. n nodes, m edges, box i of width w_i and height h_i, L fitted to the layout under test. An edge is the polyline the file stores, or the chord between centres where it stores nothing; crossings, orthogonality and node–edge separation are evaluated on the route, length and stress on the centres, and when a node moves its attachment points move with it (Supplement S3; no verdict moves on chords).

defaults to A3, so every run here passes `-align a1`). The reading direction u is the axis direction under which T_F is least for the layout as loaded, fixed before any move, the same rule as for L.

Fitting. The weights lie on the simplex; the linear program above runs over every node and direction of a corpus (100,000 to 120,000 inequalities per corpus, solved with HiGHS), and the residual is its optimum per node. Because the residual depends on the terms' scales and q does not, the pooled q under the fitted weights is reported beside it, on the fitting diagrams and on held-out diagrams under 5×2 cross-validation (five seeded halvings, each half fitted and scored on the other). The sweep, pooled q as the weight on one term grows from 0 with the rest on crossings and overlap, is the test of whether any positive weight on that term can be carried.

Statistics. The unit of inference is the diagram. Every median carries a 95% bootstrap interval over diagrams (1,000 resamples); an interval of [0.00, 0.00] or [1.00, 1.00] is a pileup at the boundary, and the share of diagrams at exactly that q is given where it matters. Hand against tool on one criterion is a Hodges–Lehmann shift with its large-sample interval, pairs that agree dropped (Supplement S5); equivalence claims are read off the sweep, never off a failure to reject. The tests are exploratory: the plan's hypotheses were fixed after a pilot (Supplement S8).

3 The corpora

Three collections of diagrams whose node coordinates a person chose, from three communities with different authoring tools, and two collections of drawings made for their own sake (Table 3); sources, versions and licences are in the repository's `data/PROVENANCE.md`.

The first three are working documents, the *hand layouts* of this paper: Reactome diagrams are laid out by curators, WikiPathways pathways are drawn in PathVisio or, from the WP5000 series on (116 of the 305), in the WikiPathways web editor, and the BPMN models are students' work of 2009 to 2019 in the Academic Initiative's Signavio editor. Nobody was asked to draw

corpus	made in	in the archive	in the band	used	median m/n
WikiPathways GPML, *H. sapiens* [33]	PathVisio, web editor	1,016	305	305	1.06
Reactome SBGN-ML [34]	curation tool	1,382	248	248	1.04
BPMN Academic Initiative [35, 36]	Signavio	29,810	6,723	300	1.11
HOLA formative drawings [29]	study tool	136	(4–13)	136	
GD Collection [21]	mixed	4,890	1,072	1,072	

Table 3: The corpora. *In the band*: largest component of 15 to 40 nodes, and for BPMN also every edge a flow or association.

nicely, and the three tools share no layout code. A diagram enters as its largest connected component of 15 to 40 nodes, container glyphs (SBGN compartments, BPMN pools and lanes) not being nodes and SBGN process glyphs being nodes. Each layout is rescaled uniformly so that its bounding box fits the unit square, box sizes with it. The BPMN corpus is a seeded random 300 of the 6,723 models in the band whose every edge is a BPMN flow or association (the archive's filtering is Supplement S6); the first 300 by model id, the sample the study was planned on, is retired to that section. Two WikiPathways files carry coordinates converted from Reactome [37] and are dropped. Deduplication by coordinates and by structure finds three pairs in the biological corpora and none in BPMN, and removing them moves no median. The 305 WikiPathways pathways carry 153 first authors. The alignment median's interval clustered by author, [0.48, 0.55], is within 0.01 of the interval over diagrams, and the per-author medians run 0.19 to 0.85 across the 20 authors with three or more diagrams.

The two further collections are drawings, not working diagrams. HOLA's formative corpus is 17 participants each laying out the same 8 abstract graphs in the study's own orthogonal editor Orthowontist, 136 drawings of 4 to 13 nodes. They are parsed from the study's published SVGs by a script in the repository, since the page carries no licence and the drawings are not redistributed here. The editor snapped bend points to a dragged node, HOLA's paper mentions no node alignment aid and no grid, and the participants started from a supplied layout. At 4 to 13 boxes the test radius is a smaller fraction of a box than in the band, and the distance-term zeros stand at radii 0.01 and 0.05 too. The GD Collection is harvested from 27 editions of the Graph Drawing proceedings, 1,072 of its figures with a largest component in the band. Its authors describe it as seeming to contain mostly manually drawn graphs and record no provenance per figure. Their audit of a random 101 papers extracted 426 graphs from 1,016 graph drawings; 69 per cent of the extracted graphs passed, 40 per cent of the vector drawings, and a pass tolerates a few missing edges. Edge directions and box sizes are absent, so only the terms needing neither are run on it.

How the coordinates were made matters for the alignment criterion. Table 4 gives the share of integer coordinates and of boxes sharing an exact coordinate with another, the figures the alignment row of Section 4 is read against. The editors' aids (a snap guide, an align command, a grid), read from their code and manuals: Signavio draws orientation lines, a guide, and snaps to them on every drag, and offers align and distribute commands (SAP Signavio Process Manager user guide, the successor of the initiative's editor). PathVisio 3 (release 3.3.0, its source) draws no guide and has no grid, and offers align-centre and stack commands on a selection. Reactome's Curator Tool (github.com/reactome/CuratorTool, 2026) draws no guide and has no grid, and offers two align commands to the first selected node. The WikiPathways web editor's aids were not read. Section 5 takes this up.

4 What the coordinates hold

Figure 2 shows one graph under the hand and three tools. Every table is from one deterministic pass over the corpora (`results/run_all.sh`); the `prism` column, the A3 kernel and the summed energies are in Supplement S1.

corpus	integer coordinates	boxes sharing an x	sharing a y	sharing either
WikiPathways	17 per cent	0.37	0.29	0.54
Reactome	67 per cent	0.06	0.12	0.17
BPMN	61 per cent	0.41	0.77	0.87

Table 4: Raw coordinates: the share that are integers, and the fraction of boxes sharing an exact coordinate with another box (`results/ties.md`).

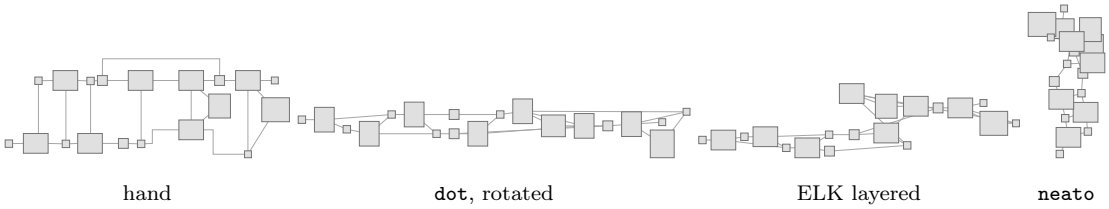


Figure 2: The same graph, the same box sizes, four layouts, one scale, the `dot` panel turned a quarter turn so its layering runs left to right: the hand drawing as stored, and the tool controls of Section 2. The hand drawing and the two layered tools share the profile the test reads: held under overlap, free under length and stress. `neato` minimises stress and is the control that gives the test its power, held at 1.00 under stress where every hand layout reads 0.00; it is run as shipped, boxes left where they overlap, and its overlap-removed variant is the `prism` control of the tables. The tool layouts carry no waypoints, so their edges are drawn as chords.

Length and stress. The median hand diagram holds no box under either term in any corpus, and 0.85 to 0.92 of diagrams sit at exactly 0.00, the [0.00, 0.00] intervals of Table 1 (`results/pileups.md`). By itself the zero says little: a smooth term is stationary at almost no point a descent did not choose, and at this radius every layout that is not a minimiser's converged output reads 0.00, the random layouts, `dot` and ELK included. Three measurements fix what the zero is worth. `neato`'s stress layouts are held at 1.00 under the same test and stay held until every box has moved about the test radius. Displaced by 0.005, 0.01, 0.02 and 0.05 of the width they read 0.94 to 0.96, 0.35 to 0.39, 0.05 to 0.06 and 0.00 (Supplement S4). So a layout at 0.00 sits more than 0.02 of the width from a minimum by the test. The hand layouts' stress value, 0.20 to 0.22, is five to twelve times the converged value (0.016 to 0.042) and under half a random layout's (0.44 to 0.51). And every diagram has improving moves (Table 5), the mean best one worth 1.3 to 1.4 per cent of the length term and 0.5 to 0.7 per cent of stress. None was taken. One such move is below perception and the departure is cumulative: a climber capped at the test radius removes 0.28 to 0.31 of $C{+}O{+}L$ and 0.12 to 0.15 of stress (Supplement S4). The hand layouts stand with the layered engines, which price neither length uniformity nor stress.

Overlap and crossings. Overlap holds every box of the median diagram at value zero, as it does in every tool layout that removes overlaps; in 5 to 18 per cent of hand diagrams some box has an overlap-lowering move, the overlap positive and rounding to 0.000 of the box area. Crossings hold every box in every layout. On BPMN only 3 per cent of diagrams offer a crossing-reducing move at all, so "crossings hold every box" is a statement about the radius and these near-trees (m/n 1.04 to 1.11), not about the author. Orthogonality reads the stored routes: its 0.73 in BPMN is Signavio's router, which draws 0.87 of edges axis-aligned at every segment, and a jitter of two units halves it (Supplement S7).

Alignment. Alignment is where the hand layout is held and `neato`'s is not. The gridiness value puts 58, 36 and 81 per cent of hand-placed boxes in an alignment of three or more. It puts none of `neato`'s there, 4 to 7 per cent of a random layout's, 3 to 9 per cent of a grid-snapped random layout's, and 88, 82 and 76 per cent of `dot`'s, since a layer is a row. That value is the

	WikiPathways			Reactome			BPMN		
term	q	dec	active	q	dec	active	q	dec	active
crossings	1.00	0.000	30%	1.00	0.000	20%	1.00	0.000	3%
overlap	1.00	0.000	18%	1.00	0.000	5%	1.00	0.000	8%
node–edge	0.88	0.047	64%	1.00	0.000	50%	1.00	0.000	12%
length	0.00	0.013	100%	0.00	0.013	100%	0.00	0.014	100%
stress	0.00	0.005	100%	0.00	0.006	100%	0.00	0.007	100%
orthogonality	0.19	0.012	100%	0.17	0.027	100%	0.73	0.008	97%
alignment A1	0.52	0.066	92%	0.21	0.069	100%	0.85	0.059	63%
flow	0.59	0.010	90%	0.62	0.012	96%	0.86	0.004	71%

Table 5: Beside the median q of the hand layouts: *dec*, the median over diagrams of the mean best decrease a single move finds as a fraction of the term's value, and *active*, the share of diagrams in which any box has an improving move. The three terms above the rule are flat at this radius; the five below are live in almost every diagram.

alignment number of this paper: it is kernel-free and does not move with the direction count. The A1 hold, 0.52, 0.21 and 0.85 against 0.06 to 0.07 by chance, is the count a first-order fit sees. It is a tie counter at the instrument's resolution: it tracks the exact-tie fractions of Section 3 to within 0.04, as a corner term should. It is also an upper bound that falls with the direction count, to 0.39, 0.07 and 0.82 at sixty-four against a chance line of 0.03, 0.03 and 0.00 (Supplement S2). Alignment does not hold the most boxes: node–edge separation holds 0.88 to 1.00 and crossings and overlap hold every box. The base energies price those three and not alignment.

Flow. Flow is the second criterion the base energies omit. It holds 0.59, 0.62 and 0.86 of hand-placed boxes against 0.28 to 0.41 by chance, and `dot` and ELK hold it at 1.00 with the term at zero: the criterion the layered engines price exactly, the person partly satisfies, tolerating a backward edge. In WikiPathways and Reactome flow holds more boxes than A1 does.

Reweighting. Table 6 is the linear program of Section 2 on four sets of terms, and Table 7 the sweep. The program is degenerate, as that section says it must be. Offered the base energy it puts all weight on the term every layout satisfies and holds 0.99 of held-out boxes. Offered all eight terms it does the same in the biological corpora, the hold on held-out diagrams swinging across WikiPathways splits as zero-residual vertices tie differently. In BPMN a vertex with 0.81 on alignment wins by the terms' scales alone and loses to crossings when the crossings column of the differences is halved. Confined to length and stress it holds 0.003 to 0.011 of the boxes. Confined to the four terms the base energies do not price it is identified in every corpus, 0.88 to 0.95 on alignment. No energy offered here makes the hand layouts minima: the best hold on held-out diagrams under those four terms is 0.65 in BPMN, 0.25 in WikiPathways and 0.15 in Reactome. The fitted four-term energy holds fewer WikiPathways boxes (0.248) than A1 alone beside crossings and overlap (0.516): the residual depends on the terms' scales, and the 0.05 of weight the fit spreads on the other three terms releases half the boxes the corner held. The identification is the corner's, not the criterion's. Under the smooth A3 kernel the same four-term fit puts 0.57 to 0.58 on flow and 0.04 to 0.08 on alignment in WikiPathways and BPMN, at a hold on held-out diagrams within 0.03 of the corner's. In Reactome it puts all weight on node–edge, with an unstable hold (Supplement S1). A term satisfied exactly by a fraction of the boxes, and possibly by the editor's aid, is a constraint the environment imposes rather than a criterion the drawer trades off. The learning-from-demonstration literature infers such constraints as their own object, the cost weights given or recovered beside them [39, 40]. The tied boxes are for A1 what every box is for overlap, a zero-residual set the program is drawn to.

terms offered	WikiPathways	Reactome	BPMN
C, O, L	overlap 1.00; residual 0.0001; held out 0.987 [0.982, 0.993]	overlap 1.00; 0.0000; 0.994 [0.990, 0.998]	crossings 1.00; 0.0005; 0.992 [0.982, 0.998]
all eight	overlap 1.00; 0.0001; 0.773 [0.235, 0.991]	overlap 1.00; 0.0000; 0.994 [0.990, 0.998]	alignment 0.81; 0.0003; 0.552 [0.479, 0.983]
L, S	stress 1.00; 0.0053; 0.003 [0.001, 0.005]	stress 1.00; 0.0055; 0.006 [0.004, 0.007]	stress 1.00; 0.0063; 0.011 [0.008, 0.014]
R, A, N, F	alignment 0.95; 0.0010; 0.248 [0.236, 0.260]	alignment 0.92; 0.0014; 0.149 [0.142, 0.157]	alignment 0.88; 0.0003; 0.651 [0.617, 0.689]

Table 6: The linear program on the simplex: the term that takes the weight (and its share where below 1), the residual per node, and the pooled q held out under 5×2 cross-validation with its range over the ten splits.

	WikiPathways		Reactome		BPMN	
term added to C+O	w=0.001	w=1	w=0.001	w=1	w=0.001	w=1
none ($w = 0$)	0.94		0.97		0.99	
length	0.008	0.004	0.010	0.007	0.012	0.010
stress	0.005	0.003	0.008	0.006	0.013	0.011
alignment A1	0.506	0.516	0.243	0.238	0.790	0.794
flow	0.576	0.606	0.613	0.625	0.811	0.819

Table 7: The sweep: pooled q, the fraction of a corpus's boxes held, as a weight w on one term joins crossings and overlap at equal weight.

The fall is a step, not a slope. Whatever weight length or stress carries, the sum holds under 0.015 of the boxes, because at almost every box some direction lowers the distance term and no other term rises there. At crossings-to-overlap ratios of 10:1 and 1:10 the same weight holds 0.005 to 0.014. Under alignment or flow the pooled hold at $w = 0.001$ equals the fraction of boxes the term holds alone, since crossings and overlap are flat at almost every box, and for the corner term that is the tie count again. Under A3 the alignment row reads 0.29, 0.13 and 0.39 at $w = 0.001$, about half the corner's figure. That is the number a learner that takes gradients of the fitted energy should expect. Axis-only moves, which preserve every perpendicular alignment tie, still improve length and stress on the median diagram of every corpus. A weight of 0.001 on length beside alignment collapses the median $C{+}O{+}A_1$ hold from 0.51, 0.22 and 0.85 to 0.11, 0.07 and 0.13. Under the constrained model of the layout literature, stress subject to alignment constraints [41, 42, 43], 0.88 to 0.90 of the median diagram's boxes still have a stress-lowering move among the directions that do not raise A1 (`results/constrained.md`, first-300 BPMN sample).

The two further collections. The same test on HOLA's 136 drawings gives crossings, overlap and node–edge 1.00, length and stress 0.00, alignment A1 0.50 with a participant-clustered interval of [0.31, 0.57] and per-participant medians from 0.14 to 0.96. On the GD Collection's 1,072 figures: crossings 1.00, length and stress 0.00 and at exactly zero in 82 and 84 per cent of figures, alignment A1 0.87. The collection's profile is the layered tools' profile, so the test cannot say whether hands or algorithms made its figures. It does say that the corpus a coordinate fit would run on first is held by the term the base energies omit and by no distance term. In the 41 to 100 node band the verdicts stand and the alignment hold falls to 0.39, 0.16 and 0.74 (Supplement S7).

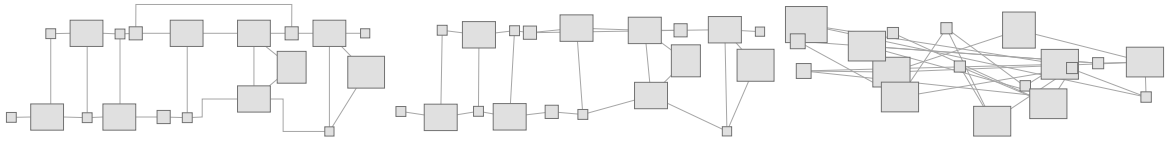


Figure 3: The grid alternative of Section 5: the grid alone does not produce the alignment hold. One BPMN model, 19 boxes; the corpus medians of the three panels are 0.85, 0.19 and 0.07. Left, the drawing as its author left it. Centre, every box displaced by up to 15 units, three times the $0.19\,d$ past which the directional test resolves a misalignment. Right, every box placed uniformly at random in the same bounding box and snapped to the same grid: the grid is available and the author is not. Box sizes are unchanged throughout. The hand panel draws the routes the file stores; the two control panels draw chords, their boxes no longer being where the routes ran.

5 What else could produce this

The editor snapped to a grid, so alignment is the tool's. Excluded. The detected pitch is one unit in Reactome and BPMN and none in WikiPathways, where exact alignment is twice as common as integer coordinates. A random layout snapped to the drawing's unit grid (pitch one where none is detected), with the same boxes and the same bounding box and no author, holds 0.06 to 0.09 over three seeds in every corpus (Figure 3). Snapped to the plausible working pitch of five units it holds 0.076 (first-300 BPMN). The grid is available to the random layout and does not produce the effect. Displacing every hand- placed box by up to half a pitch leaves the hold within 0.03 of where it was. That test cannot fail: the corner releases a box only once its offset from its nearest agreement exceeds $0.19\,d$ (Supplement S2), 3 to 5 units against a half pitch of 0.5. Displacing past that takes the hand layouts to the snapped random level (0.06 at a half-width of 120 units; Supplement S7).

The editor aligned the boxes, by a snap guide or an align command. Not separable on these corpora, and conceded: a grid control cannot see either. The hold is highest, 0.85, in the one editor with a drag guide, Signavio. The other two editors draw no guide, and there the exact ties came from the hand or from an align command applied to a selection. WikiPathways holds 0.52 and Reactome 0.21 under align commands of the same kind, so the aid does not fix the level. The hold appears under four editors that share no code, HOLA's Orthowontist included at 0.50, and is flat across four generations of WikiPathways pathway numbers (0.47 to 0.53, Supplement S7), so it is not one editor's artefact. Separating the hand from the align command needs an interaction log, and these corpora carry none. Marriott et al. [44] bound the other side: in a recall task node alignment was among the features participants did not preserve, so the alignment these coordinates hold is a property of how the layouts were made, not evidence that anyone perceived it. Every claim here is about the coordinates as data for a fit, and for a fit the attribution changes nothing: the coordinates hold a term the energy lacks either way.

The lanes place the boxes. Of BPMN boxes, 82 per cent sit in a lane or pool, and a horizontal lane constrains y. Of the held boxes, 28 per cent are held by a column, which no horizontal lane imposes, 14 per cent by a row shared across lanes, and 58 per cent by a row within one lane. Those lanes are a median 6.3 box heights tall, so the row inside them was chosen.

The rest, each with the test that excluded it, are Table 8.

alternative	test	result
Centre distances on box diagrams	border-to-border distances, L refitted	hand 0.00 in every corpus; `neato`, which minimised the centre form, not held under the border form (Supplement S3)
The radius and the directions	$d \in \{0.005, 0.01, 0.02, 0.05\}$; 8 to 64 directions	hand verdicts the same at every d and count; the A1 level falls with the direction count and the gap to the control does not (Supplement S2)
Near-trees make overlap and crossings easy	q against m/n over diagrams	crossings track density by hand and under `dot` alike; length and stress are 0.00 in the sparse and the dense half both (`results/sparsity.md`)

Table 8: The remaining alternatives, each with the test that excluded it.

6 Related work

Aesthetic criteria and their weighting go back to Tamassia, Di Battista and Batini [1] and were made an explicit energy by Davidson and Harel [2]. Purchase [3] gave the criteria their metrics, among them node orthogonality, the fit of node positions to an imaginary grid. Bennett et al.'s survey [45] lists it and uniform edge length side by side, each with a Gestalt rationale and, as of 2007, neither evaluated. The test here is such an evaluation, at the coordinates, and it separates the two. Table 9 is what was known, by how it was asked.

The learned quality scores [20, 53, 54] and Barbosa and Barreto's genetic algorithm, which co-evolved a weight vector against a ranking users supplied [55], sit with Masui and Klammler: none tests whether an example is a minimum of the learned score. Alignment as a human criterion has been known since 2012 in graph drawing [48] and since 1985 in drawing editors. There Pavlidis and Van Wyk's beautifier inferred alignment constraints from the near-alignments of a rough drawing [56], Igarashi et al. made the inference interactive [57], and the guides and align commands of Section 5 are their descendants. Energies and constraint systems that price it exist: HOLA's gridiness step [29], O'Donovan's alignment terms [12], and stress under alignment constraints [41, 42, 43]. Van Wageningen, Mchedlidze and Telea [58] show drawings with identical metric values that differ in quality, so a fitted energy with no residual would still not be the energy people use. The claims here stay on the negative side. Marriott et al.'s test of feature preservation against random redrawings [44] is the nearest precedent in layout to the nulls used here.

7 What follows

The charge against the energies is bounded twice. `dot` and ELK hold alignment at 0.42 to 1.00 without a term for it, since a layer puts its nodes on one line and Brandes– Köpf straightens edges across layers, so the missing term convicts the energies and not the layered engines. And the energies that do price alignment, HOLA's, O'Donovan's and the constraint systems, are not the ones a coordinate fit to a graph-drawing corpus would be offered.

For a tool that learns its weights. Offer alignment among the terms, and flow where the graph is directed, or the fit learns nothing of what the drawer did. That is an addition: edge length and stress keep their weight, since a term inert at human coordinates is not thereby unwanted, and the preference evidence runs the other way [51, 47]. People prefer the layouts a stress minimiser makes, do not draw that way, and in recall do not preserve the alignment they do draw [44]. No study yet takes production and perception from one corpus. An energy fitted to coordinates a snap guide or an align command produced learns the aid: an editor that already snaps gains a term it enforces on every drag, and a layout engine without a guide gains what the

work	what it gives, and what it leaves
By having people draw	
van Ham, Rogowitz [46]	71 of 73 hand layouts raise edge-length variance above the force-directed optimum half of them started from
Dwyer et al. [47]	32 participants drew 48 layouts of one graph; 188 online judges chose among 9 of them and 3 automatic layouts, and their choices track stress; the user layouts carry 3 to 9 times the force-directed stress in that paper's Table 2
Purchase, Pilcher, Plimmer [48]	drawing from adjacency lists: crossing removal the strongest aesthetic, alignment to a grid important
Kieffer et al. [29]	17 participants' layouts of 8 graphs ranked by 66 judges: gridiness correlates with rank on all 8, the source of HOLA's alignment step
Marriott et al. [44]	25 participants redrawing small diagrams from memory preserve symmetry, collinearity and orthogonality, and not node alignment
Bernstein, Soffer [49, 50]	layout features of human-made process models named and measured, reading direction and alignment among them; no fit
Mooney et al. [51]	people prefer low stress and find shortest paths better as it falls, graphs to 50 nodes
Mooney, Hegemann, Wolff, Wybrow, Purchase [21]	4,890 figures harvested from these proceedings, provenance not per figure; measured in Section 4
By learning from examples	
Masui [14]; Spönemann et al. [15]; Klammler et al. [16]; Zhang et al. [17]	constraint functions from good and bad layouts; weights from selections, from good-against-worsened pairs, from 64,222 preference labels; none tests whether an example is a minimum of the learned energy, and none fits human coordinates
SmartGD [18]; DeepDrawing [19]; Kwon et al. [20]	coordinates imitated, of algorithm layouts selected by metric, 68 to 81 per cent of them `neato`'s in four of SmartGD's seven collections; SmartGD replaces its examples as it finds better ones, so it does not take them as optimal
Yuan et al. [52]	users' sub-layouts merged into one drawing as target distances of a Laplacian-constrained embedding, weights fixed; coordinates as constraints, nothing learned
Flud [22]	annealing continued from crowd layouts of pathway diagrams, the closest prior descent from a hand layout, without a cap, a reference or a per-term reading
O'Donovan, Agarwala, Hertzmann [12]	a 122-parameter energy over single-page designs fitted by nonlinear inverse optimisation, each example taken as optimal: the assumption examined here; their term set contains alignment

Table 9: What is known, by how it was asked.

aid gives. The number to report for a fitted energy is the fraction of a held-out drawing's boxes the energy holds under the terms whose median value at the demonstrations is above zero. That is $q_{0.02}$ at sixteen directions with the fitted L, with the decrease and active columns beside it. A bare held fraction is bought by weighting a term everyone satisfies, as the degenerate fit of Section 4 shows. For the corner term the exact-tie share of Table 4 is the ceiling. Any tool that emits `dot` with positions can report the number: `data/dot2corpus.py drawing.gv my.txt`, then `station direct -corpus my.txt -weights 0,0,0,0,0,1,0,0 -align a1`. The gain is small and measured. When each box is replaced in turn against the frozen rest, alignment does not move the median distance to the drawing's position, and it raises the share of boxes landing on the drawing's row or column by a fifth to a half (`results/reinsert.md`). A preference or likelihood fit reads a different quantity. Hand layouts carry half the stress of random layouts of the same graphs and more length non-uniformity (0.22 to 0.28 against 0.19 to 0.22), so such a fit would weight stress and not length uniformity, where the coordinate fit weights neither. For

a layered engine, whose parameters are spacings and placement strategies rather than weights, the result is a target level: on BPMN 81 per cent of hand-placed boxes sit in an alignment of three or more and 72 per cent of ELK's, on Reactome 36 against 69. The levers are the node placement strategy and the straight-edge preference, and neither targets an alignment fraction.

Limits. The alignment count is specific to box-and-arrow diagrams of 4 to 100 boxes drawn in editors; it falls with size; its level is stated at sixteen directions. Which unpriced criterion the four-term fit names depends on the kernel offered. The negative result separates hand layouts from converged ones and from nothing else; its content is the distance and the step. The term set is the force-directed family's. Untested: a minimum node gap (which every layered engine prices, and which is live where overlap is degenerate), total edge length, compactness, bends, symmetry and the edge alignment of unequal boxes. Whether a layout started from an automatic one the person then edited, the production case for a shipped editor, is not recorded. Reactome's Curator Tool has a hierarchical layout command, so its diagrams may have, and HOLA's participants started from a supplied layout. The tool controls are laid out without the containers the person worked inside, and three of their eight terms are read on chords. The per-drawer spread of Sections 3 and 4 says a population weight may be the wrong object. The BPMN authors' courses are not recorded, so diagrams from one assignment may not be independent. The plan the study started from is beside the paper, its outcomes in Supplement S8.

Declaration. The author has no funding and no competing interest to declare; every human-made layout here is secondary use of published data.

References

[1] R. Tamassia, G. Di Battista, C. Batini. Automatic Graph Drawing and Readability of Diagrams. *IEEE Transactions on Systems, Man, and Cybernetics* 18(1):61–79, 1988.

[2] R. Davidson, D. Harel. Drawing Graphs Nicely Using Simulated Annealing. *ACM Transactions on Graphics* 15(4):301–331, 1996.

[3] H. C. Purchase. Metrics for Graph Drawing Aesthetics. *Journal of Visual Languages and Computing* 13(5):501–516, 2002.

[4] S. G. Kobourov. Force-Directed Drawing Algorithms. In *Handbook of Graph Drawing and Visualization*, CRC Press, 2013.

[5] R. K. Ahuja, J. B. Orlin. Inverse Optimization. *Operations Research* 49(5):771–783, 2001.

[6] A. Keshavarz, Y. Wang, S. Boyd. Imputing a Convex Objective Function. *IEEE International Symposium on Intelligent Control*, 613–619, 2011.

[7] T. C. Y. Chan, T. Lee, D. Terekhov. Inverse Optimization: Closed-Form Solutions, Geometry, and Goodness of Fit. *Management Science* 65(3):1115–1135, 2019.

[8] P. Englert, N. A. Vien, M. Toussaint. Inverse KKT: Learning Cost Functions of Manipulation Tasks from Demonstrations. *International Journal of Robotics Research* 36(13–14):1474–1488, 2017.

[9] B. Berret, E. Chiovetto, F. Nori, T. Pozzo. Evidence for Composite Cost Functions in Arm Movement Planning: An Inverse Optimal Control Approach. *PLoS Computational Biology* 7(10):e1002183, 2011.

[10] A. Y. Ng, S. Russell. Algorithms for Inverse Reinforcement Learning. In *Proc. 17th International Conference on Machine Learning* (ICML 2000), Morgan Kaufmann, 663–670, 2000.

[11] S. Levine, V. Koltun. Continuous Inverse Optimal Control with Locally Optimal Examples. In *Proc. 29th International Conference on Machine Learning* (ICML 2012), Omnipress, 41–48, 2012.

[12] P. O'Donovan, A. Agarwala, A. Hertzmann. Learning Layouts for Single-Page Graphic Designs. *IEEE Transactions on Visualization and Computer Graphics* 20(8):1200–1213, 2014.

[13] R.-T. Chen, K. Xue, C. Gao, Y. Shi, T. Xu, P. Xie, S. Xu, M. Yuan, C. Qian, Z.-H. Zhou. How Can Reinforcement Learning Achieve Expert-level Placement? In *Proc. 63rd ACM/IEEE Design Automation Conference* (DAC 2026), arXiv:2604.25191, 2026.

[14] T. Masui. Evolutionary Learning of Graph Layout Constraints from Examples. *Proc. UIST '94*, 103–108, 1994.

[15] M. Spönemann, B. Duderstadt, R. von Hanxleden. Evolutionary Meta Layout of Graphs. *Diagrams 2014*, LNCS, 16–30, 2014.

[16] M. Klammler, T. Mchedlidze, A. Pak. Aesthetic Discrimination of Graph Layouts. *Journal of Graph Algorithms and Applications* 23(3):525–552, 2019.

[17] P. Zhang, X. Li, X. Wang, H.-W. Shen, Y. Hu. Beauty in the Eye of AI: Aligning LLMs and Vision Models with Human Aesthetics in Network Visualization. arXiv:2604.03417, 2026.

[18] X. Wang, K. Yen, Y. Hu, H.-W. Shen. SmartGD: A GAN-Based Graph Drawing Framework for Diverse Aesthetic Goals. *IEEE Transactions on Visualization and Computer Graphics* 30(8):5666–5678, 2024.

[19] Y. Wang, Z. Jin, Q. Wang, W. Cui, T. Ma, H. Qu. DeepDrawing: A Deep Learning Approach to Graph Drawing. *IEEE Transactions on Visualization and Computer Graphics* 26(1):676–686, 2020.

[20] O.-H. Kwon, T. Crnovrsanin, K.-L. Ma. What Would a Graph Look Like in This Layout? A Machine Learning Approach to Large Graph Visualization. *IEEE Transactions on Visualization and Computer Graphics* 24(1):478–488, 2018.

[21] G. J. Mooney, T. Hegemann, A. Wolff, M. Wybrow, H. C. Purchase. Universal Quality Metrics for Graph Drawings: Which Graphs Excite Us Most? In *33rd International Symposium on Graph Drawing and Network Visualization* (GD 2025), LIPIcs 357, 30:1–30:20, 2025. Collection at `doi:10.4230/artifacts.25065`, commit `d113537`, CC BY-SA 4.0.

[22] A. Bharadwaj, D. Gwizdala, Y. Kim, K. Luther, T. M. Murali. Flud: A Hybrid Crowd-Algorithm Approach for Visualizing Biological Networks. *ACM Transactions on Computer-Human Interaction* 29(1), 2022.

[23] N. D. Ratliff, J. A. Bagnell, M. A. Zinkevich. Maximum Margin Planning. In *Proc. 23rd International Conference on Machine Learning* (ICML 2006), 729–736, 2006.

[24] B. D. Ziebart, A. Maas, J. A. Bagnell, A. K. Dey. Maximum Entropy Inverse Reinforcement Learning. In *Proc. 23rd AAAI Conference on Artificial Intelligence* (AAAI 2008), 1433–1438, 2008.

[25] A. Aswani, Z.-J. M. Shen, A. Siddiq. Inverse Optimization with Noisy Data. *Operations Research* 66(3):870–892, 2018.

[26] K. Smelser, J. Miller, S. Kobourov. "Normalized Stress" is Not Normalized: How to Interpret Stress Correctly. *2024 IEEE Evaluation and Beyond: Methodological Approaches for Visualization* (BELIV), 41–50, 2024.

[27] R. Ahmed, C. Erten, S. Kobourov, J. Lotz, J. Miller, H. Taraz. Size Should Not Matter: Evaluating Network Visualizations with Stress. *arXiv preprint arXiv:2408.04688*, 2024.

[28] C. D. Schulze, M. Spönemann, R. von Hanxleden. Drawing Layered Graphs with Port Constraints. *Journal of Visual Languages and Computing* 25(2):89–106, 2014.

[29] S. Kieffer, T. Dwyer, K. Marriott, M. Wybrow. HOLA: Human-like Orthogonal Network Layout. *IEEE Transactions on Visualization and Computer Graphics* 22(1):349–358, 2016.

[30] R. Ahmed, F. De Luca, S. Devkota, S. Kobourov, M. Li. Graph Drawing via Gradient Descent, $(GD)^2$. *Graph Drawing (GD 2020)*, LNCS, 3–17, 2020.

[31] T. Kamada, S. Kawai. An Algorithm for Drawing General Undirected Graphs. *Information Processing Letters* 31:7–15, 1989.

[32] E. R. Gansner, Y. Koren, S. C. North. Graph Drawing by Stress Majorization. *Graph Drawing (GD 2004)*, LNCS, 239–250, 2005.

[33] A. Agrawal, H. Balcı, H. Hanspers et al. WikiPathways 2024: Next Generation Pathway Database. *Nucleic Acids Research* 52(D1):D679–D689, 2024. GPML release 20260810, *Homo sapiens*, CC0.

[34] A. Fabregat et al. Reactome Diagram Viewer: Data Structures and Strategies to Boost Performance. *Bioinformatics* 34(7):1208–1214, 2018. Human SBGN-ML diagrams of release 97, CC0 (`https://reactome.org/license`).

[35] M. Kunze, A. Luebbe, M. Weidlich, M. Weske. Towards Understanding Process Modeling: The Case of the BPM Academic Initiative. *BPMN 2011*, LNBIP, 44–58, 2011.

[36] M. Weske, G. Decker, M. Dumas, M. La Rosa, J. Mendling, H. A. Reijers. Model Collection of the Business Process Management Academic Initiative, version BPMAI-29-10-2019. Zenodo, 2020, `doi:10.5281/zenodo.3758705`, CC BY 3.0.

[37] A. Bohler et al. Reactome from a WikiPathways Perspective. *PLOS Computational Biology* 12(5):e1004941, 2016.

[38] G. Di Battista, A. Garg, G. Liotta, R. Tamassia, E. Tassinari, F. Vargiu. An Experimental Comparison of Four Graph Drawing Algorithms. *Computational Geometry* 7(5–6):303–325, 1997.

[39] G. Chou, N. Ozay, D. Berenson. Learning Constraints From Locally-Optimal Demonstrations Under Cost Function Uncertainty. *IEEE Robotics and Automation Letters* 5(2):3682–3690, 2020.

[40] D. R. R. Scobee, S. S. Sastry. Maximum Likelihood Constraint Inference for Inverse Reinforcement Learning. In *International Conference on Learning Representations* (ICLR 2020), OpenReview.net, 2020.

[41] T. Dwyer, Y. Koren, K. Marriott. IPSep-CoLa: An Incremental Procedure for Separation Constraint Layout of Graphs. *IEEE Transactions on Visualization and Computer Graphics* 12(5):821–828, 2006.

[42] T. Dwyer, K. Marriott, M. Wybrow. Dunnart: A Constraint-Based Network Diagram Authoring Tool. *Graph Drawing (GD 2008)*, LNCS 5417, 420–431, 2009.

[43] J. Hoffswell, A. Borning, J. Heer. SetCoLa: High-Level Constraints for Graph Layout. *Computer Graphics Forum* 37(3):537–548, 2018.

[44] K. Marriott, H. Purchase, M. Wybrow, C. Goncu. Memorability of Visual Features in Network Diagrams. *IEEE Transactions on Visualization and Computer Graphics* 18(12):2477–2485, 2012.

[45] C. Bennett, J. Ryall, L. Spalteholz, A. Gooch. The Aesthetics of Graph Visualization. *Computational Aesthetics in Graphics, Visualization, and Imaging* (CAe 2007), Eurographics, 57–64, 2007.

[46] F. van Ham, B. Rogowitz. Perceptual Organization in User-Generated Graph Layouts. *IEEE Transactions on Visualization and Computer Graphics* 14(6):1333–1339, 2008.

[47] T. Dwyer, B. Lee, D. Fisher, K. I. Quinn, P. Isenberg, G. Robertson, C. North. A Comparison of User-Generated and Automatic Graph Layouts. *IEEE Transactions on Visualization and Computer Graphics* 15(6):961–968, 2009.

[48] H. C. Purchase, C. Pilcher, B. Plimmer. Graph Drawing Aesthetics: Created by Users, Not Algorithms. *IEEE Transactions on Visualization and Computer Graphics* 18(1):81–92, 2012.

[49] V. Bernstein, P. Soffer. Identifying and Quantifying Visual Layout Features of Business Process Models. In *Enterprise, Business-Process and Information Systems Modeling* (BPMDS/EMMSAD 2015), LNBIP 214, 200–213, 2015.

[50] A. Burattin, V. Bernstein, M. Neurauter, P. Soffer, B. Weber. Detection and Quantification of Flow Consistency in Business Process Models. arXiv:1602.02992, 2016.

[51] G. J. Mooney, J. Miller, M. Wybrow, S. Kobourov, H. C. Purchase. Stress in Graph Drawings: Perception, Preference, and Performance. In *33rd International Symposium on Graph Drawing and Network Visualization* (GD 2025), LIPIcs 357, 38:1–38:23, 2025.

[52] X. Yuan, L. Che, Y. Hu, X. Zhang. Intelligent Graph Layout Using Many Users' Input. *IEEE Transactions on Visualization and Computer Graphics* 18(12):2699–2708, 2012.

[53] H. Haleem, Y. Wang, A. Puri, S. Wadhwa, H. Qu. Evaluating the Readability of Force Directed Graph Layouts: A Deep Learning Approach. *IEEE Computer Graphics and Applications* 39(4):40–53, 2019.

[54] W. Huang, M. L. Huang, C.-C. Lin. Evaluating Overall Quality of Graph Visualizations Based on Aesthetics Aggregation. *Information Sciences* 330:444–454, 2016.

[55] H. J. C. Barbosa, A. M. S. Barreto. An Interactive Genetic Algorithm with Co-evolution of Weights for Multiobjective Problems. In *Proc. Genetic and Evolutionary Computation Conference* (GECCO 2001), 203–210, 2001.

[56] T. Pavlidis, C. J. Van Wyk. An Automatic Beautifier for Drawings and Illustrations. *SIGGRAPH '85*, 225–234, 1985.

[57] T. Igarashi, S. Matsuoka, S. Kawachiya, H. Tanaka. Interactive Beautification: A Technique for Rapid Geometric Design. *UIST '97*, 105–114, 1997.

[58] S. van Wageningen, T. Mchedlidze, A. Telea. Same Quality Metrics, Different Graph Drawings. In *33rd International Symposium on Graph Drawing and Network Visualization* (GD 2025), LIPIcs 357, 7:1–7:13, 2025.